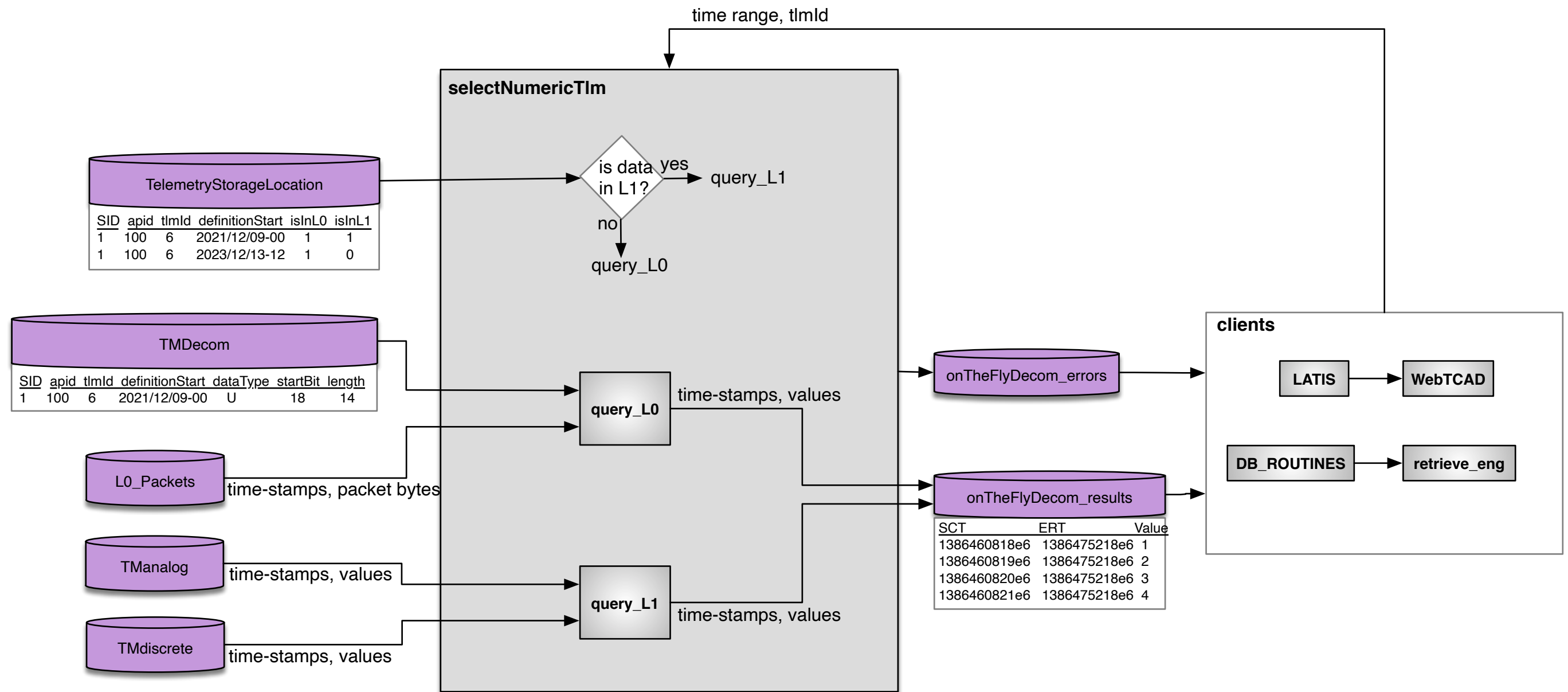


On-the-fly Decom



- IXPE Example:
1. The user selects telemetry item IXPE P0100SCNT (tImId=6) from 2023/12/13 to 2023/12/14, and specifies SID=1 (FLIGHT).
 2. The client application calls the selectNumericTIm stored procedure, passing the SID, tImId and time-range.
 3. selectNumericTIm queries the TelemetryStorageLocation table with this tImId and time-range, and determines that data before 2023/12/13-12:00 is stored in L1, and data after that is only stored in L0.
 4. selectNumericTIm calls query_L1 from 2023/12/13-00:00 to 2023/12/13-12:00. From the dataType it knows to query TManalog. It queries TManalog_SID1, and stores the results in the onTheFlyDecom_results table.
 5. selectNumericTIm calls query_L0 from 2023/12/13-12:00 to 2023/12/14-00:00. This queries the TMDecom table over the time range, and finds there's only one row, so only one query is necessary for the entire time range. From the startBit and length it knows to query for 2 bytes of each packet, starting 2 bytes into the packet. It queries L0_Packets_SID1, extracts a 14-bit unsigned integer from those 2 bytes of each packet, and adds the results to the onTheFlyDecom_results table.
 6. The stored procedure returns, and the client application queries the onTheFlyDecom_results table to get the data. It also queries the onTheFlyDecom_errors table, and if not empty, displays the error strings to the user.